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Summary

With over 15 years of experience in programming, operating and testing SaaS software and storage systems; I'm keen to exploit my knowledge further, helping with local AI inference (RAG) driven applications

WORK EXPERIENCE

Santevet | Online bookings

France - remote 🇫🇷

23 - till present

Key Responsibilities

- Devops – build orchestration tools for **RAG applications** (local LLM)
  - Build tools with local LLM with Elixir and Nx.
  - RAG applications include LLM code review.
- Enforce the software robustness of the Phoenix online bookings SaaS
  - Fix several critical bugs like this one found in the commanded dependency.
  - Fix aws instance sizes and wrong linux settings.
  - Solving some front bugs.
- Bring observability:
  - Add high resolution metrics (user metrics).
  - Add monitoring with Prometheus / PromQL.
  - Build salient Grafana Dashboards.
- Stripe: add a card payment method to Santevet products and migrate our Stripe api.
- Implement a garbage collector for CQRS SQL data.

Github

- Commanded (CQRS/ES): fix an OOM bug present for years in the commanded dependency;
- Prometheus high resolution metrics : create a small memory footprint prometheus counter able to track different activities for all users, used intensively by our Analytics
- local llm scripts

Environment

- AWS - Postgres - Phoenix - CQRS/ES - EventStore - Commanded - ElasticSearch - Stripe API - PromQL - Broadway - Rust

Motorola Solutions

Glostrup, Denmark - on-site 🇩🇰

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Key Responsibilities

- Devops for a cloud-based telecom product that connects Land Mobile Radio systems over the internet;
- **Reliable software programming - kubernetes devops**
- Software: write new kubernetes microservices (service-level requirement, ..), associated tests and deployed the services in all regions;
  The role included tasks like writing / running the deployment pipelines, the CI/CD integration, templates, helm chart, monitoring, logging and observability
- SRE tasks: successfully implement **a SLA measurement** microservice, join on-call rotation, run post-mortem, improve reliability

Environment

- Kubernetes - Helm - Docker - Elixir - Elk - Azure Pipelines - Telemetry - PagerDuty - Prometheus - Continuous Integration - Rabbitmq - Grafana - Redis - Phoenix - GraphQL.

Kuantic | Lead

Sofia-Antipolis, France

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Key Responsibilities

- Fleet Management SaaS – Technical interface for Stellantis
- Architectural tasks:
  - Modernizing the software development and deployment process
  - Backend code refactoring
  - **Reshape the fleet management API into a more general data broker API** (concepts + software)
- Reliability: improve the stack reliability and extensibility with the help of a MQ
- Big-data: Transform a fleet trafic model into a data broker business model: mapReduce pipelines, configurable customers queues, third-party export pipelines
- Tooling: add tools like Elastic Search and Elixir Dashboards
- Troubleshoot some issues
- Performance: improve overall performance with some caching
- Side project: develop and maintain a SaaS based on Phoenix / LiveView

Github

- A local key-value cache;
- A nif cache for openstreetmap.

Environment

- NoSQL · MapReduce · Erlang · Distributed Systems · RabbitMQ · Broadway · Elixir · SQL · Docker · Kubernetes · Phoenix · Elastic Search

Scaleway | R&D

Paris, France

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Key Responsibilities

- Distributed Cloud Storage – R&D – Devops
  - From scaleway requirements, define a distributed and fault-tolerant storage architecture
  - Implement a flexible, lightweight protocol layer (nbd) in erlang (where it shines), allowing for example to relocate customers storage while in use
  - *Linux, Live release upgrade, app profiling, riak LRU cache, block dependency graph, users volume live relocation, riak node crash recovery, data flow performance analysis, commodity computing, webmachine rest api*
- **AI fraud detection system**
  - Ownership of the fraud (rent servers abuses) decision engine design
  - Build the features and python workflow (SQLAlchemy, Rest API)
  - The built AI fraud detection is used in production and is fully automated for the support team, probably the first AI agent ever used in the company
  - *Scikit-learn, Python, SQLAlchemy, Postgresql*
- Online SAS san servers as IOT (POC) for live snapshots and statistics data collection
  topics: qemu-nbd, iot
- Erlang distributed testing of AWS s3
  topics: basho_bench, tsung

Github

- a spatio-temporal logger.

Environment

- Scikit-Learn · PostgreSQL · NoSQL · Storage Area Network (SAN) · Machine Learning · Erlang · Python - SQLAlchemy - Distributed Systems · Quickcheck · Docker · Qemu · Linux Kernel

Meetic, +500 people

Paris, France

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Key Responsibilities

- Distributed chat system – main backend developer
- Maintainer of the messaging stack (Erlang/ Ejabberd **+200K simultaneous users**)
- Architect and code new requirements for the Meetic chat system
- Add the xmpp protocol bosh extension to the backend
- Package, benchmark and R&D work in erlang:
  - Development of a riak-jabberd roster (contacts)
  - Development of a game framework as a cluster extension for Ejabberd to support simultaneously ten of thousands of human interactions stored in Riak
  - A realtime 3D supertracker to monitor fraudulent chat messages
  - Benchmark RabbitMQ for messages logging
- Side project: rewrite in Erlang a C++ biological follicular/ovarian growth simulator I built for INRA, (ATT/C++ coroutines), my first project in Erlang ❤️
- Side projects: open-source a real-time extension for erlang, allowing to schedule hard real-time messages, illustrated with two N2O projects:
  - An Erlang multi-player midi sequencer
  - An interactive physics simulator

Github

- A real-time NIF extension for Erlang, allowing to schedule real-time messages.

Consultant

embedded software

Paris, France

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- **Embedded programming – Philips / LG / Sagem mobiles**
  - Based on the 18Crypt profile, architect of the NXP mobileTV CAS feature (conditional access); this multi-site project involved people from the Netherlands, India and Ireland (S3)
  - Feature leader of Puma phone media manager, specify & code the real-time requirements (apollo, js)
  - Development of a new video player automaton in C for Sagem Mobile
  - Build a host-based verification system tool to ensure the middleware correctness, this tool uses logic programming techniques on a host to drive the tests on the embedded system
  - Android devices: build a verification tool to test low-level Android java hooks (thread, java fields / methods, state machines)
- **Embedded programming – IPTV STB**
  - Add multi-threading (EMM/ECM and Tuner) for the Philips STB DLI (Device Layer Interface); develop a porting layer for PVR services based on ST40 crypto security
  - As a field engineer at BYTEL for the BBOX IPTV Product, investigates Technicolor stb Middleware
  - Rewrite in Erlang my massively parallel C++ coroutines simulator of biological processes (based on Cox Renewal Theory statistics model)
- **DVB-H Mobile TV – Orange**

- Head of the developments for the next generation SIM & KDA conditional access Architecture of a poc (2 patents) server - mobile - card , mp4 encryption schemes, licence packaging
- Manage the relationships with major smart card suppliers, video software vendors and third-party software companies
- Semi-formal testing of the sim software using logic programming techniques , customize VLC to decode the rtp h264 ismacryp stream

CP8 Transac

Louveciennes, France

July 97 - June 2002

- **Project leader of CP8 first javacard (3 patents including 2 US)**
  - Did the coordination between INRIA, the mask and the CP8 R&D teams. This project included a virtual machine simulator along with other tools like secure loader or bytecode optimization programs and benchmarks suite software
  - R&D: wrote many javabytecode optimizations and javacard demos like **a mastermind solver** for benchmark
  - Replace the old proprietary VB tools with web applications for the new HTTP smartcard interface (one patent)
  - Technical consultancy for a French bank (Caisse d'Epargne, B0' application)
- **Secure protocols for e-commerce, Carte Bleue**
  - Port the French banking protocol (B0') for javacard
  - Add **multi-threading programming for the SimToolkit** (3 patents have been filled)
  - Proof of concept of a sms-based E-purse transaction system (Proton smartcard and server)
  - Open-source the Prolog PC/SC interface , used to run semi-formal tests in logic programming

EDUCATION

Groupe Ecoles Centrales

Centrale Méditerranée, France

- Graduated in Computer Engineering
  Minor: Industrial IT and embedded systems
- Competitive classes preparing students to the best engineering schools ("Les grandes Ecoles")

Computer SKILLS

RAG development:	Elixir, Nx, Bumblelebee, ollama, langchain and Jina
Systems:	SaaS, Kubernetes / Docker, local LLM, Azure pipelines CI/CD, low latency Erlang/OTP, Elixir ecosystem, Phoenix, Tailwind-css, github actions, riak (nosql), linux kernel (bloc storage), graphQL, sqlite, Postgresql, SQLAlchemy, Tsung, redis, C++ Coroutines
Prediction - Observability:	RAG - local LLM programming - PromQL - High Resolution Counters
Programming languages:	Erlang / Elixir, C/C++, Python, JavaScript prototype, java-bytecode, SQL, ...
Methodologies:	DevOps best practices (Motorola)
Protocols:	Nbd, aws, xmpp, iptv, TR-069, mpeg-4 streaming (RFC 3984), ISMACryp 1.0, Etsi-Tetra
Embedded:	Linux, Nucleus, RTKI, VME, JavaCard and Android
Smart Cards:	EMV, SIM Card, B0' Banking Application, E-Purse, OpenPlatform

Personal PROJECTS

Prometheus:	Create a small memory footprint telemetry_metrics_prometheus_core counter able to track millions of different counters
LiveView-based SaaS:	Develop and maintain a SaaS based on Phoenix / LiveView / live SVG
SuperIs:	A multi-volumes, files tokenizer, indexer and search engine
Caching systems:	Small libraries on top of a NOSQL database
Spatio-temporal logger:	A tiny library to debug concurrency in distributed systems, both in time and space
Hardware clock:	ALSA-hardware-based clock in C (NIF) for real-time message processing in Erlang
Ejabberd Game framework:	Framework for multi-player games on top of Riak and Ejabberd